System Design Document: Cymonic Content Factory

1. Executive Summary & Architecture Overview

The Cymonic Content Factory is an autonomous, human-in-the-loop (HITL) marketing pipeline designed to eliminate the manual bottleneck between product engineering and go-to-market execution.

Instead of a basic LLM wrapper, the application utilizes a sequential **4-Agent Cascade** orchestrated via CrewAI. The architecture strictly separates concerns: extraction (Research Agent), generation (Copywriter Agent), validation (Editor Agent), and visual synthesis (Visual Director Agent). By forcing the AI to extract a structured "Source of Truth" first, downstream generation agents are constrained to factual data, significantly reducing hallucination risks.

2. The 4-Agent Pipeline

The pipeline operates sequentially, passing contextual memory from one node to the next:

- **Node 1: The Research Agent:** Ingests messy, unstructured data (via raw text or web scraping). Its sole purpose is to output a highly structured Markdown fact-sheet. Crucially, it identifies and isolates ambiguous claims into a "Red Flags" category.
- **Node 2: The Copywriter Agent:** Acts as the creative engine. It autonomously classifies the product as B2B or B2C, routing social media to the correct platforms (e.g., banning Instagram for enterprise software). It generates a dynamically formatted blog post and two A/B tested email variants (Logic-driven vs. Emotion-driven).
- **Node 3: The Editor Agent (Hallucination Gate):** Operates entirely as a validation layer. It cross-references the Copywriter's draft against the Researcher's "Red Flags" list, actively scrubbing unverified or exaggerated claims before rendering the final copy.
- **Node 4: The Visual Director Agent:** Analyzes the final, polished copy to engineer a hyper-specific image generation prompt.

3. Tech Stack & Strategic Trade-Offs

- **LLM Engine: Groq API (Llama 3.3 70B)**
    - *Decision:* Chose Groq's LPU-accelerated inference over standard OpenAI GPT-4o.
    - *Trade-off:* While GPT-4 has a slight edge in highly complex reasoning, Groq provides near-instantaneous token generation at $0.00 compute cost. For a marketing factory that requires rapid, iterative agent-to-agent communication,

high velocity and sustainable unit economics heavily outweighed the marginal reasoning deficit.

- **Frontend Framework: Streamlit**
    - *Decision:* Chose Streamlit over a traditional React/Next.js stack for rapid Python-native deployment.
    - *Trade-off:* Streamlit is notoriously rigid regarding UI customization and component state lifecycles. This was mitigated by utilizing heavy CSS injection to override default components (achieving a premium B2B SaaS aesthetic) and engineering a custom `st.session_state` "memory vault."
- **Image Generation: Pollinations.ai (Flux Model)**
    - *Decision:* Chose Pollinations via direct URL fetching.
    - *Trade-off:* Bypassed the need for complex API authentication or heavy local model hosting. The trade-off is a strict 10-image-per-hour rate limit, which is handled gracefully by alerting the user directly in the UI.

4. Problem-Solving & Edge Case Handling

- **Context Window / TPM Rate Limiting:** Web scraping often pulls in massive amounts of hidden HTML, triggering Groq's 12,000 Tokens-Per-Minute (TPM) limit. The BeautifulSoup4 scraper is engineered to automatically strip navigation/script tags and heavily truncate the text to 2,500 characters, ensuring the multi-agent cascade never crashes mid-execution.
- **Widget Garbage Collection (UI Persistence):** Streamlit aggressively deletes widget states upon reruns. To allow users to edit the AI's output without losing their data, a custom `on_change` callback was built to intercept user keystrokes and lock them into an immutable `st.session_state` vault before Streamlit's garbage collection triggers.
- **AI Typography Failures:** Image generators struggle with exact spelling. The Visual Director's prompt engineering explicitly instructs the AI to strip special characters, shorten the brand name, and render the text "subtly" and "small" to mask minor structural generation errors.

5. UI/UX Innovation & "The Extra Mile"

Beyond the core brief, the application was engineered for enterprise usability:

- **Human-in-the-Loop (HITL) Editor:** Fully autonomous publishing is a liability for enterprise brands. A custom toggle switch transforms the read-only markdown into a live, editable text area, providing a mandatory human review layer before exporting the `.md` file.
- **Execution Analytics:** A dynamic dashboard tracking agent assembly speed, node count, and compute costs. This highlights an acute awareness of cloud computing economics, proving the architecture is scalable and cost-effective.
- **A/B Testing Native Integration:** Modern marketing relies on data, not guesses. The system natively generates multi-variant copy tailored to different psychological buyer profiles, matching real-world marketing workflows.